CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Faculty of Technology and Engineering
B. Tech. (IT) Sem. – IV University Examination, 2010-11
IT207: Computer Networking-I

Date: 11.05.11, Wednesday Time: 10:00a.m. To 01:00p.m. Maximum Marks: 70

Instructions:
(1) All questions are compulsory.
(2) Indicate clearly, the option you attempt along with its respective question Number.
(3) Figures to right indicate marks.
(4) Rough work is done in the last page of main supplementary; don't write anything on the question paper.

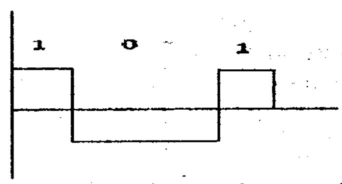
SECTION-I

Q-1 Answer the following questions. **[11]**

[A] State True/False. [1]
1. ICMP Protocol works at Network Layer
2. DNS works at Transport Layer

[B] Why effectiveness of data communication does depend on "Accuracy"? [1]

[C] State True/False. [1]
1. Following is a Periodic Digital Signal.



2. To decompose a composite signal to various periodic signal components, Galvan analysis is needed.

[D] What are ~~peer-to-peer processes with respect to OSI model?~~ [1]
[E] What is the purpose of carrier signal in modulation? [1]
[F] What are DTE and DCE? Give example of DTE and DCE devices. [1]
[G] What is an intelligent modem? [1]
[H] What king of information can be obtained from constellation diagram? [1]
[I] What is bit rate and baud rate of digital signal? [1]
[J] Performance of Network also depends on number of current users. Why is it so? [1]
[K] Why are modems needed for telephonic internet communication? [1]

Q-2 Answer the following questions. **[12]**

[A] Topology refers to the way a network is laid out. What are the topologies of computer network? Write down at least two disadvantages of any one topology. [4]
[B] Phase shift Keying is used to convert digital signal to analog signal. Explain it with constellation diagram. [4]
[C] Write down electrical, mechanical and functional characteristics of EIA-232 standard. [4]

OR

[A] What are the functions of Physical Layer, Data Link Layer, Network layer and Application Layer? [4]
[B] What are the two types of Biphase digital encoding? By taking suitable example, draw and explain waveform generation in it. [4]
[C] What are the differences between synchronous and asynchronous transmission? [4]

Q-3 Answer the following questions.(Attempt any Four) **[12]**

[A] Write down the differences between OSI model and TCP/IP model.

[B] By taking suitable example, explain HDB3 digital encoding technique.

[C] In dial-up modem, why the maximum data rate in the uploading direction is 33.6 kbps and data rate in the downloading is 56 kbps.

[D] What is null modem? Why is it required? Draw the pin-connection to make null modem.

[E] By taking suitable example, explain 8-QAM.

SECTION-II

Q-4 Answer the following questions. **[11]**

[A] State True/False. [1]

1. Flow Control mechanism is related to traffic on network.
2. "Who should send now?" is solved by Line Discipline.

[B] What is collision? [1]

[C] What is critical angle with respect to optical communication? [1]

[D] What are the disadvantages of packet switching network? [1]

[E] What kind of error is undetected by checksum? [1]

[F] What are the types and subtypes of packet switching approach? [1]

[G] What is frequency range of signal that can be carried over coaxial cable? [1]

[H] What are the uses of BSC control frames? [1]

[I] Why is slot reservation needed in DQDB? [1]

[J] What are the light sources for optical cable? [1]

[K] Error detection usually done in the ________ layer of OSI model. [1]

Q-5 Answer the following questions. **[12]**

[A] What is structure of co-axial cable? How does it differ from twisted pair cable? [4]

[B] With the help of diagram, explain internal working of TDM bus. [4]

[C] By taking suitable example, explain checksum mechanism for error detection. [4]

OR

[A] What are the two types of frame format in Binary Synchronous Communication? Explain frame format. [4]

[B] What are the advantages and disadvantages of Optical fiber cable? [4]

[C] Draw and Explain the fields of Ethernet Frame Format. [4]

Q-6 Answer the following questions.(Attempt any Four) **[12]**

[A] What are the station types, configuration types and modes of communication in HDLC?

[B] What is the mechanism of poll/select method in line discipline?

[C] In transmission of data on media, imperfection cause impairment in the signal sent through the medium. What are the types of impairment? How does it impact quality of signal?

[D] Explain Following Terms:

1. 10Base2
2. 10Base5
3. 10BaseT
4. 100BaseT
5. 100BaseFx
6. 100BaseT4

[E] Describe the analog hierarchy in which group of signals is successively multiplexed onto higher bandwidth lines.